

Why new future markers are often banned from negative contexts

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Abstract Some languages have future markers that are banned from or dispreferred in negative contexts. This results in situations of asymmetry in which certain grammatical distinctions made in the affirmative are unavailable in the negative. We propose a diachronic explanation for such asymmetries: future markers first grammaticalize in affirmative contexts and only later, if at all, spread to negative contexts, sometimes with a time lag of hundreds of years. During the period where a novel future marker occurs in the affirmative but is incompatible with negation, the synchronic situation is perforce one of asymmetry. We further argue that newly grammaticalized future markers tend to be banned from negative contexts due to their semantics and discourse function. New futures are typically immediate futures. We suggest that speakers conventionally use novel immediate futures to make their statements more informative. As a consequence, these future markers require the assertion to be stronger than its alternatives. In affirmative contexts, the use of an immediate future gives rise to a strong assertion. On the other hand, in negative contexts, the use of an immediate future leads to a weak assertion, which conflicts with the conventional discourse function of these markers. As a result, novel immediate futures tend to be restricted to affirmative contexts, i.e., they are positive polarity items.

1 Introduction

Negative constructions can be divided into two basic types: symmetric and asymmetric (Miestamo 2005). A symmetric negative construction is one in which the negative

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is structurally identical to the corresponding affirmative, except for the presence of the negative marker(s), as demonstrated by (1) in Lezgian. An asymmetric negative construction is one in which there are further structural differences between the negative and the affirmative, as in English (2), where the negation of simple past and present tense verbs involves a *do* auxiliary and a non-finite verb. Asymmetric negation is frequent in the world's languages: Miestamo (2005: 171) reports on 107 languages with asymmetric standard negation out of a worldwide sample of 179 languages.

- (1) Lezgian (Nakh-Dagestanian, Lezgetic; Haspelmath 1993: 127, 245, cited in Miestamo 2005: 10)
- a. xürünwi-jri ada-waj meslät-ar ħaču-zwa.
villager-PL.ERG he-ADEL advice-PL take-IPFV
'The villagers take advice from him.'
 - b. xürünwi-jri ada-waj meslät-ar ħaču-zwa-č.
villager-PL.ERG he-ADEL advice-PL take-IPFV-NEG
'The villagers do not take advice from him.'
- (2) a. Mary came.
b. Mary didn't come.

A very common type of asymmetric negation is paradigmatic asymmetry, where certain grammatical distinctions made in the affirmative are unavailable in the negative or, more rarely, vice versa. For instance, in some languages, certain tense-aspect-mood (TAM) markers are restricted to affirmative contexts, as in Burmese (3), where the presence of the negative marker precludes any TAM marking. Thus, Burmese instantiates both constructional asymmetry (the negative lacks a TAM suffix) and paradigmatic asymmetry (certain grammatical distinctions are neutralized in the negative).¹

- (3) Burmese (Sino-Tibetan; Cornyn 1944: 1213, cited in Miestamo 2005: 123)
- a. θwâ-dé
go-ACT
'(he) goes, went'

¹ Another type of paradigmatic asymmetry, which is not the focus of this paper, is one where the same grammatical distinctions are available in the negative, but the language employs morphologically distinct markers in affirmative and negative contexts (Miestamo 2005: 116-118). For instance, Tera (i) uses a dedicated perfective marker in the negative.

- (i) Tera (Afroasiatic, Chadic; Newman 1970: 128, 14, cited in Miestamo 2005: 116)
- a. ali wà masa koro.
PN PFV buy donkey
'Ali bought a donkey.'
 - b. ali n`ə masa koro ɓa.
PN PFV buy donkey NEG
'Ali didn't buy a donkey.'

- b. $\theta w\hat{a}$ -mé
go-POT
‘(he) will go’
- c. $\theta w\hat{a}$ -bí
go-PRF
‘(he) has gone’
- d. ma- $\theta w\hat{a}$ -bû
NEG-go-NEG
‘(he) does/did/will not go, has not gone’

Givón (1978: 97-99) proposes a diachronic explanation for synchronic situations of paradigmatic asymmetry: novel tense-aspect markers first grammaticalize in affirmative contexts and only later spread to negative contexts. During the period where a novel tense-aspect marker occurs in the affirmative but is incompatible with negation, the synchronic situation is perforce one of paradigmatic asymmetry.

This paper investigates paradigmatic asymmetry in future marking. Following Givón (1978), we suggest that this asymmetry often arises from the grammaticalization of a new future marker. This is illustrated by Ancient Egyptian, a language with an exceptionally long recorded history. In Middle Egyptian, the classical stage of the Ancient Egyptian language, a synthetic verb form is used to express future reference (4-a), and this form can be negated (4-b). During the same period, a new future construction, derived from an allative preposition, emerges in the affirmative (4-c) but is incompatible with negation (4-d).² It is only in the Late Egyptian stage of the language that the new future construction becomes compatible with negation (5-b) (Grossman & Polis 2014: 54). Thus, the grammaticalization of the new future leads to a temporary yet extended state of paradigmatic asymmetry. Symmetry is eventually restored once the limitations on the distribution of the new future are lifted.

- (4) Middle Egyptian (Afroasiatic, Egyptian; Grossman & Polis 2014)
- a. **ms-s**
give.birth:FUT-3SG.F
‘She will give birth.’ (old future)
 - b. **nn ms-s r-nḥḥ**
NEG give.birth:FUT-3SG.F for-eternity
‘She will not give birth.’ (old future)
 - c. **iw dpt r-iy-t m-ḥnw**
AUX boat to-come-INF from-home
‘A ship will come from home.’ (new future)
 - d. ***nn iw dpt r-iy-t m-ḥnw**
NEG AUX boat to-come-INF from-home
‘A ship will come from home.’ (new future)

² We have marked this example with an asterisk to indicate that it is ungrammatical, but more conservatively, we can say that examples like this are not found in Early Egyptian corpora.

- (5) Late Egyptian (Afroasiatic, Egyptian; Grossman & Polis 2014)
- a. iw-f r-tȝyt-t r wʿ-mʿ
 AUX-3SG.M to-take-2SG.F to INDF-place
 ‘He will take you to a place.’ (new future)
- b. yʒ bn iw-sn r-rḥ fȝy-f
 indeed NEG AUX-3PL to-know bring-3SG.M
 ‘Indeed, they will not be in the position to contest it.’ (new future)

As Miestamo (2005: 220-221) points out, it is yet to be explained *why* new markers tend to grammaticalize first in the affirmative and only later in the negative. To address this gap, we propose a semantic-pragmatic explanation for the tendency of future markers to be restricted to affirmative contexts in the early stages of their grammaticalization, thereby supplying the missing piece of the puzzle in Givón’s proposal. We particularly focus on novel future markers that express temporal imminence, which is a common function of novel future markers, according to Bybee et al. (1994: 271-273).

In affirmative contexts, a statement with an immediate future is logically stronger than an alternative statement with a simple future. For example, ‘The coffee is about to run out (in the near future)’ asymmetrically entails ‘The coffee will run out (some time in the future)’. By contrast, in negative contexts, an immediate future has a weakening effect: ‘The coffee is not about to run out (in the near future)’ is asymmetrically entailed by ‘The coffee will not run out (at any given time in the future)’.

We suggest that speakers employ the strengthening effect of immediate futures to make their statements more informative. This creates a statistical bias: these markers are almost exclusively used in affirmative contexts, where they strengthen the assertion. In other words, new futures tend to be banned from negative contexts because of their semantics and discourse function. Over time, this statistical tendency might become conventionalized, in which case the strengthening effect becomes part of the conventional discourse function of the future marker. Ultimately, the distributional constraints on novel futures may either be relaxed, thereby reinstating paradigmatic symmetry, or become grammatically obligatory. This paper presents historical evidence supporting both scenarios.

We further note that immediate future markers that are restricted to affirmative contexts have much in common with lexical positive polarity items (PPIs) expressing imminence such as *The train will arrive **any time now***. Such temporal adverbs are likewise infelicitous in negative contexts since using them in a context where they weaken the overall meaning conflicts with their conventional discourse function. We propose an exhaustification-based analysis of imminence PPIs, which applies both to lexical PPIs and immediate future markers that behave like PPIs.

In sum, our account suggests a recurrent cause of paradigmatic asymmetry in future marking. However, it is by no means exhaustive: there are other types of paradigmatic asymmetry in future marking (see Section 2), which would require a different explanation than the one we propose in this paper. It remains to be seen whether the gist of our account can be extended to some of these cases. That is,

it is possible that a newly-grammaticalized future marker whose primary function is not an immediate future would nonetheless be conventionally associated with a strengthening effect, resulting in a distributional restriction to affirmative contexts. In fact, Bybee et al. (1994: 235) contend that speakers' desire to make stronger statements is one of the functional motivations behind grammatical innovations: "one of the reasons new periphrases grammaticize is that they express in a stronger fashion the meaning that has eroded and weakened in the older gram." However, exploring this possibility is left for future research.

This paper is structured as follows. Section 2 illustrates several cases of paradigmatic asymmetry in future marking and proposes a diachronic pathway of a transition from paradigmatic asymmetry to a restoration of symmetry, which is exemplified by Ancient Egyptian. This section also offers an overview of the types of grammatical distinctions that tend to be neutralized in negative contexts and discusses the typical lexical sources and semantics of future markers that are restricted to affirmative contexts. Section 3 proposes a formal analysis of novel immediate future markers as PPIs. In Section 4, we speculate on how a PPI future marker may lose its polarity sensitivity and become compatible with negation, resulting in paradigmatic symmetry. Section 5 discusses previous approaches to paradigmatic asymmetry in TAM marking. Section 6 concludes.

2 Paradigmatic asymmetry in future marking

This section presents cross-linguistic evidence suggesting that newly-grammaticalized future markers are sometimes (and perhaps typically) incompatible with negation. The data are based on several cases of paradigmatic asymmetry reported in the literature. At this point, we are unable to estimate whether and to what degree these cases represent a frequent cross-linguistic pattern and leave it for future research to quantify its prevalence.

2.1 Diachronic pathway

Following Givón (1978: 97-99), we suggest that paradigmatic asymmetry in future marking often arises from the grammaticalization of a new future marker, though there are exceptions, as we point out later in this subsection. That is, negation tends to be a conservative context that does not readily admit novel future markers. In the time it takes for the novel marker to spread to negative contexts (if it does so at all), the synchronic situation is one of asymmetry. Thus, the grammaticalization of the novel future marker tends to proceed through one of two diachronic paths: (i) the novel future marker gradually spreads to negative contexts, thereby restoring paradigmatic symmetry; or (ii) the future marker's incompatibility with negation

becomes grammatically obligatory. We outline this recurrent diachronic pathway in (6), noting that a novel future marker may become “stuck” at an intermediate stage.

(6) *Diachronic pathway of development of paradigmatic symmetry in future marking:*

- In **Stage 1**, the new future is restricted to affirmative contexts, while an older future (or another construction that can have future reference) may occur in both affirmative and negative contexts.
- In **Stage 2**, occurrences of the new future in negative contexts are sporadic and rare, and the older future form is preferred in negative contexts.
- In **Stage 3**, the new future becomes wholly compatible with negation.

The full pathway in (6) has been exemplified by Ancient Egyptian (4)-(5), where occurrences of the new future construction in negative contexts are attested well after this construction was conventionalized (Grossman & Polis 2014: 54).

Synchronic situations of paradigmatic asymmetry correspond to Stages 1 and 2 in (6). Tigré (Afroasiatic) instantiates Stage 1 asymmetry: the new future circumfix (ə)gə(l)-...-tu is restricted to affirmative contexts, and the older imperfective is used with negation (7).³

(7) Tigré (Afroasiatic, Semitic; Noam Faust, p.c.)

- a. aze sala:di bǝ-na, la:kin sala:di-na fajr gə-tə-tmam-tu.
now money with-1PL but money-1PL tomorrow FUT-3SG.F-end-FUT
‘Now we have money, but tomorrow our money will run out.’
- b. aze sala:di bǝ-na, la:kin sala:di-na fajr yi-tə-tamam-ni.
now money with-1PL but money-1PL tomorrow NEG-3SG.F-end-NEG
‘Now we have money, but tomorrow our money will not run out.’

In other cases of Stage 1 asymmetry, a novel construction can mark future reference in affirmative contexts, whereas in negative contexts, only the original lexical meaning of the source construction is available. In Tok Pisin (8), the auxiliary *laik* marks immediate future, but in negative sentences, it can only mean ‘want’—its lexical source meaning (Verhaar 1995: 149).

(8) Tok Pisin (English-lexified creole; Verhaar 1995: 150-151)

- a. strongpela win na ren i laik i kam.
strong wind and rain PM want PR come
‘[When there is lightning, this shows us that] there will soon be wind gusts and heavy rain.’
- b. em i no laik kaikai.
3SG PM NEG want eat
‘He doesn’t want to eat at all.’ Not:
(Intended: ‘He is not going to eat at all.’)

³ The novel future construction gə-...-tu is derived from the allative preposition [?]əgə(l) and the copula *tu* ‘it is’ (Grossman & Polis 2014: 38), whereas the older construction is based on the Semitic imperfective, as in many other Semitic languages (Lipiński 1997: 357).

In the Abu Ghosh dialect of Palestinian Arabic, the novel immediate future marker *da* ‘he is about to’ is restricted to affirmative contexts (9-a)-(9-b), whereas the simple future marker *rah* ‘be going to’ is compatible with negation (9-c)-(9-d).⁴ In accordance with the pathway in (6), the future marker that co-occurs with negation appears to be diachronically older.⁵ This results in paradigmatic asymmetry since the remoteness distinction is neutralized in negative contexts. Similarly to Tok Pisin (8), only the lexical meaning of *da*- ‘he wants’ is available in negative contexts (Nadine Abdel-Rahman, p.c.).

- (9) Palestinian Arabic, Abu Ghosh dialect (Afroasiatic, Semitic; Nadine Abdel-Rahman, p.c.)
- a. el-kahwa da-txalles[§].
DEF-coffee want.PRS.3SG.M-run.out.IPFV
‘The coffee is about to run out.’
 - b. ?el-kahwa de-ʃ txalles[§].
DEF-coffee want.PRS.3SG.M-NEG run.out.IPFV
(Intended: ‘The coffee is not about to run out.’)
 - c. el-kahwa rah txalles[§].
DEF-coffee FUT run.out.IPFV
‘The coffee is going to run out.’
 - d. el-kahwa meʃ rah txalles[§].
DEF-coffee NEG FUT run.out.IPFV
‘The coffee is not going to run out.’

In Stage 2 asymmetry, the novel future construction is strongly dispreferred in negative contexts but not completely banned. For instance, in Quebec French (10), the newer periphrastic future (PF) comprises 73% of the total number of future reference utterances in the spoken Ottawa-Hull French Corpus, while the older synthetic future

⁴ By simple future, we mean a future marker that does not encode remoteness distinctions. Hence, this term refers to both inflectional and periphrastic future markers.

⁵ The prefix *da*- is a phonologically reduced form of the volition verb *bidd*- ‘want’. For instance, *da*- ‘he is about to’ is derived from *bidda* ‘he wants’, and *na*- ‘we are about to’ is derived from *biddna* ‘we want’ (Nadine Abdel-Rahman, p.c.). Future markers that have grammaticalized from *bidd*- are found throughout the Levantine area, and they express various types of future meanings (Leddy-Cecere 2018: 57-58). In the Abu Ghosh dialect, represented by the data in (9), only the reduced form has a future meaning, while the unreduced form *bidd*- only has the original lexical meaning. However, in other dialects of Palestinian Arabic, *bidd*- only occurs in an unreduced form and has the meaning of a simple future (AbuAmsha 2020). Note that there is considerable controversy about whether *bidd*- is also the lexical source of the highly-grammaticalized *b*- future marker, which has a wide geographic distribution, or whether *b*- has a different lexical source (Owens 2018: 217-219). In any case, *da*- is clearly distinct from *b*-. We tentatively suggest that the *da*- immediate future is probably younger than the simple future marker *rah* (9-c), which is derived from a motion verb (*rayih* ‘go’). This conclusion mostly rests on the fact that *da*- and *bidd*- have a more limited geographical distribution compared to *rah*, which is found in many Arabic varieties from Iraq to Algiers (Leddy-Cecere 2020: 611). In Egyptian Arabic, the initial stages of the grammaticalization of *rah* are attested as early as the 16th and 17th centuries (Leddy-Cecere 2020: 606). We do not know when the grammaticalization of *bidd*- and *da*- began, but the literature does not seem to report on early attestations of *bidd*- as a future marker.

(SF) comprises 20% of the utterances. However, in negative contexts, the synthetic future is used 97% of the time (Poplack & Dion 2009: 572-573). Thus, we suggest that a categorical (Stage 1) or statistical (Stage 2) restriction to affirmative contexts might represent different stages of the same diachronic pathway.

- (10) Quebec French (Indo-European, Romance; Poplack & Dion 2009: 574)
 ... *il va virer* (PF) *la clef*, *il va la rebarrer* (PF) *puis va la redébarrer* (PF).
Il s'apercevra (SF) *pas qu'elle était débarré*.
 'He is going to turn the key, he is going to lock it again, then going to unlock it. He won't notice that it was unlocked.'

The conservativeness of negative contexts with respect to future marking can be further demonstrated by a comparative study of 20 Northwestern South-Central Trans-Himalayan languages (Sino-Tibetan) by Konnerth (2020). Konnerth has found that these languages are highly innovative in developing future markers, but the negative future is typically expressed by /ni/, which is derived from an equational copula and can be reconstructed as an affirmative future marker in the ancestor language. Thus, these languages typically have multiple affirmative future markers, which are derived from various sources—direct speech, purpose marking, desiderative, and allative constructions—but usually employ an “old” future form in the negative.

It is important to note, however, that the grammaticalization of a future marker does not necessarily proceed to Stages 2 and 3. There may be cases where the incompatibility of a future marker with negation is stable. The Telkepe variety of Northeastern Neo-Aramaic presents an interesting case. The old future form *b-*, which is derived from a volition verb, is incompatible with negation (11-a)-(11-b), and the present indicative form is used instead in the negative (Coghill 2010: 368).⁶ In other words, the *b-* future remained at Stage 1 of the pathway in (6). In contrast, the novel future construction *zi(l)*, which is derived from a motion verb, can be negated (11-c)-(11-d). Thus, Telkepe Neo-Aramaic is an apparent counter-example to the cross-linguistic pattern we have proposed since the new future can be negated while the old future cannot. However, Coghill (2010: 382, 386) suggests that *zi(l)* was originally banned from negative contexts as well but later became negatable. This is supported by the fact that other dialects of Neo-Aramaic do not allow the negation of *zi(l)* as an independent verb, and in the closely-related Alqosh dialect, *zi(l)* is altogether banned from negative contexts both as an independent verb and as an auxiliary. Hence, the novel future construction *zi(l)* seems to have proceeded through the three stages in (6) in the Telkepe dialect, bypassing the older *b-* future construction, which remained at the first stage.

- (11) Telkepe Neo-Aramaic (Afroasiatic, Semitic; Coghill 2010: 368)
 a. *b-pāθəx*.
 FUT-open.3M
 'It will open.'

⁶ The present indicative can only have future reference in negative contexts (Coghill 2010: 368).

- b. *la-b-pāθəx.
NEG-FUT-open.3M
(Intended: ‘It will not open.’)
- c. zi-lə pāθəx.
PROSP-L.3M open.3M
‘It’s going to open.’
- d. lá-zzi-lə mbāšəl.
NEG-PROSP-L.3M cook.3M
‘He is not going to cook.’

Similarly, the Kvanda variety of Bagvalal (Nakh-Daghestanian), as described by Kibrik et al. (2001), has a synthetic future *-s* that cannot be negated, while a morphologically unrelated synthetic negative future *-a:č’e* can be used to express future reference in negative clauses (see Footnote 1 for a discussion of this type of paradigmatic asymmetry).⁷ The latter is historically derived from the imperfective stem, and it appears to be an “old present”, as evidenced by the fact that it can still have present reference with certain stative verbs. Thus, it appears that both synthetic futures, the affirmative *-s* and the negative *-a:č’e*, are old forms. By contrast, the newer periphrastic future, which consists of a future participle and a copula, can be negated. Thus, this language seems to be another case where the older future form has remained at Stage 1 without proceeding to the latter stages in (6).⁸ Therefore, we suggest that the proposed diachronic pathway in (6) can lead to complex situations of paradigmatic asymmetry since languages may have multiple future markers at different stages of development. In Tekelpe Neo-Aramaic and Kvanda Bagvalal, older future markers became “stuck” at Stage 1, while newer futures have apparently proceeded to later stages.

Finally, it should be noted that not every instance of paradigmatic asymmetry in future marking aligns with the diachronic pathway we have outlined. For instance, in Oji-Cree (Algonquian), the novel standard negation marker *ci-*, which is derived from an irrealis subordinating prefix, cannot co-occur with future tense verbs, while the older standard negation marker *hsii-*, which dates back to Proto-Algonquian, is compatible with all tenses (Windsor 2022). Thus, in this language, paradigmatic asymmetry in future marking arises from the grammaticalization of a novel negative marker rather than a novel future marker, unlike the other languages discussed in this paper.

⁷ A similar situation is found in Ladakhi (Sino-Tibetan), where the future auxiliary is omitted in the negative, but there is a special negative prefix that expresses future reference (Koshal 1979: 244, cited in Miestamo 2005).

⁸ We thank Timur Maisak (p.c.) for bringing this phenomenon to our attention.

2.2 Neutralization of grammatical distinctions

In this subsection, we address the following question: what grammatical distinctions are neutralized in situations of paradigmatic asymmetry in future marking? It seems that in many cases, paradigmatic asymmetry affects remoteness distinctions. According to Bybee et al. (1994: 271-273), novel future constructions are often immediate futures. It appears that such markers tend to be banned from negative contexts. In our data, the immediate futures in the Abu Ghosh dialect of Palestinian Arabic (9) and Tok Pisin (8) cannot be negated, and a simple future is used instead in the negative. Similarly, in Mongsen Ao (Sino-Tibetan), the newly-grammaticalized immediate future suffix is incompatible with negation, and the irrealis suffix is used instead in negative contexts (Coupe 2013: 1121-1122), as illustrated in (12).

(12) Mongsen Ao (Sino-Tibetan; Coupe 2013: 1119, 1122)

- a. nì nə m̀ŋsən-əɿ hjutsə sa-ɿ̀.
1SG AGT Mongsen-ANOM story say-IMM
'I'll tell a Mongsen man's story.'
- b. nì nə hjutsə à m̀ə-sa-ì(-ù?).
1SG AGT story one NEG-say-IRR(-DECL)
'I won't tell a story.'

In other languages, a remote future is likewise incompatible with negation. For example, Nungon (Nuclear Trans New Guinea) distinguishes between the near future and the remote future. This distinction is neutralized under negation, and the irrealis form is used instead of either of the future forms (Sarvasy 2017: 271). A similar situation is found in the Urama dialect of Kiwai (Kiwaian), where the distinction between the near future and the remote future is lost in the negative, and a dedicated negative future marker is employed (Brown et al. 2016: 35-37; cited in Peck et al. 2020).

(13) Nungon (Nuclear Trans New Guinea; Sarvasy 2017: 271)

- a. na-wang-ka-t
eat-PROB.SG-NF-1SG
'I will eat it (later today)'
- b. na-i-t-ma
eat-IRR.SG-1SG-RF
'I will eat it (tomorrow or beyond)'
- c. ma=na-i-t
NEG-eat-IRR.SG-1SG
'I will not eat it (later today or ever)'

However, there are also cases where it is not remoteness that is neutralized. In Bambara (Niger-Congo), the Assertive Future marker *ná / nà* expresses certainty, while the Intentional Future *bénà* is mostly used when speaking of plans or events in the near future. The Assertive Future cannot be negated, whereas the Intentional Future has a negative counterpart *ténà*. Thus, negation neutralizes a modal distinction in

this language (Idiatov 2000). In Bumthang (Sino-Tibetan), a distinction is made between personal and impersonal verb forms. In negative clauses with future reference (expressed by irrealis), only the impersonal form can be used (Peck et al. 2020: 356).

(14) Bumthang (Sino-Tibetan; Peck et al. 2020: 355-356)

- a. ngat yamba chogor-o gae-sang.
1SG tomorrow Chamkhar-ALL go-PERS.IRR
'I will go to Chamkhar tomorrow.'
- b. gon yamba chogor-o gae.
3SG tomorrow Chamkhar-ALL go-IMPERS.IRR
'She will go to Chamkhar tomorrow.'
- c. yamba chogor-o mé-gae.
tomorrow Chamkhar-ALL NEG-go-IMPERS.IRR
'(I/She) will not go to Chamkhar tomorrow.'

Finally, in some languages, the distinction between future and non-future is altogether lost in the negative, as was demonstrated in Burmese (3). In Hup (Naduhup), the novel future suffix *-teg* cannot co-occur with negation. In negative contexts, future reference is expressed by the inchoative aspect marker *-ay* (15-a), which can also have present reference in this context (15-b) (Epps 2008a,b).

(15) Hup (Naduhup; Epps 2008b: 535)

- a. ?āh ham-níh-ay
1SG go-NEG-INCH
'I'm not going to go.'
- b. pe?-níh-ay
sick-NEG-INCH
'I'm not sick.'

2.3 Lexical sources and semantics

One might wonder whether there is a causal relation between the lexical source of a novel future construction and its compatibility with negation. While we are not able to offer a conclusive answer to this question, we observe that the future markers in our data that are incompatible with negation come from a variety of lexical sources, as shown in Table 1.

Given the diversity of lexical sources, a different question arises: are novel future markers always restricted to affirmative contexts in the earlier stages of their grammaticalization? It seems that the answer is negative. For instance, there seems to be no evidence for an early stage in the grammaticalization of English *be going to* where it was altogether banned from negative contexts. The literature places the initial grammaticalization of *be going to* in the seventeenth century (Petré 2016: 122, a.o). While we are not aware of a study that investigated the use of *be going to* in negative contexts as it was developing into a future marker, one can find pretty

Table 1 Lexical sources of future markers that are incompatible with negation

Lexical source	Language	Future marker	Reference
Volition	Palestinian Arabic	<i>da-</i>	Nadine Abdel- Rahman (p.c.)
	Tok Pisin	<i>laik</i>	Verhaar (1995)
	Telkepe Neo-Aramaic	<i>b-</i>	Coghill (2010)
Motion verb	Quebec French	<i>aller</i>	Poplack & Dion (2009)
	Bambara	<i>ná / nà</i>	Idiatov (2000)
	Alqosh Neo-Aramaic	<i>zi(l)</i>	Coghill (2010)
Allative	Tigré	<i>(ə)gə(l)-... -tu</i>	Raz (1983)
	Middle Egyptian	<i>r-</i>	Grossman & Polis (2014)
Purposive	Hup	<i>-teg</i>	Epps (2008a)
	Mongsen Ao	<i>-uù</i>	Coupe (2013)

early examples of *be going to* in negative contexts. For example, in (16), *be going to* expresses intention, and in (17) it expresses imminence—in neither case is there an implication of motion. Nonetheless, it might turn out that such instances were relatively rare in that period, corresponding to Stage 2 in the pathway in (6). At any rate, being restricted to affirmative contexts does not seem to be a necessary stage in the grammaticalization of a future marker.

- (16) I am not going to say that every strange Birth, none excepted, was conceived under ♄ [astronomical symbols for Saturn and Jupiter]; but I say the Contingency is so *frequent*, that, it may be, it deserves to be noted by those who understand better. [John Goad, *Astro-Meteorologia*, 1686]⁹
- (17) Husband: Why, then, I will tell you; indeed, I am not going to break, and I hope I am in no danger of it, at least not yet.
 ...
 Wife: Well, but let us talk a little more seriously of it; you are not going to break, that is, not just now, not yet, you said; but, my dear, if it is then not just at hand, but may happen, or is in view at some distance, may not some steps be taken to prevent it for the present, and to save us from it at last too. [Daniel Defoe, *The complete English tradesman*, 1726]¹⁰

Instead of looking at the lexical source of a newly-grammaticalized future marker, we suggest that it might be more telling to consider its semantics and discourse function. In many of the cases discussed in the paper, we cannot determine what these functions were because a future marker might have undergone considerable semantic change since the initial stages of its grammaticalization. However, we can focus on a particular function frequently associated with novel future markers in various languages: immediate future. According to Bybee et al. (1994: 271-273), there are both formal and semantic indications that immediate futures are typically

⁹ Goad, John. 1686. *Astro-Meteorologia: Or aphorisms and discourses of the bodies caelestial, their natures and influences*. London: Printed by J. Rawlins for Obadiah Blagrave.

¹⁰ Defoe, Daniel. 1726. *The complete English tradesman*. London: Charles Rivington.

3 Novel immediate futures as positive polarity items

(18) a. The train is going to arrive any time now.
b. *The train isn't going to arrive any time now.

An immediate future implies that the event in question is imminent. This can be shown by its incompatibility with temporal adverbs that refer to the non-near future. For example, Palestinian Arabic *da-* can co-occur with a temporal adverb expressing imminence, but not with one relating to a distant time period (19).

- (19) Palestinian Arabic, Abu Ghosh dialect (Afroasiatic, Semitic; Nadine Abdel-Rahman, p.c.)
- a. el-ħavila da-tu:s^fal oxra xames daka:jk.
DEF-package want.PRS.3SG.M-arrive.IPFV other five minutes
'The package is about to arrive in five minutes.'
- b. #el-ħavila da-tu:s^fal eʃfaher eɕḡəj:
DEF-package want.PRS.3SG.M-arrive.IPFV DEF.month coming

¹² It's worth noting that a PPI can be a complex expression: the PPI *any time now* consists of *any*, which by itself is a negative polarity item (NPI), and *time* and *now*, which are not polarity sensitive by themselves.

(Intended: ‘The package is going to arrive next month.’)

We observe that temporal adverbs expressing imminence tend to be PPIs, as demonstrated in (20). Given that immediate future markers share the imminence meaning component with these PPI adverbs, it is not surprising that they have a similar distribution and are restricted to affirmative contexts.

- (20) a. I’ll be back {in no time / in a sec / in a jiffy / in two shakes (of a lamb’s tail) / before you can say Jack Robinson / before you know it}.
- b. *I won’t be back {in no time / in a sec / in a jiffy / in two shakes (of a lamb’s tail) / before you can say Jack Robinson / before you know it}.

A notable property of PPIs is that they are not altogether banned from negative contexts (Baker 1970; Szabolcsi 2004). For example, *any time now* cannot co-occur with negation in a main, declarative clause (18-b), but it can do so in a negative question (21). In general, PPIs tend to be compatible with negation in sentences where the entire clause is embedded within an environment where NPIs are licensed, such as a question (21) or a conditional, a phenomenon known as “rescuing”. Furthermore, a PPI can appear in the scope of negation when the sentence is an echo denial (sometimes referred to as emphatic denial): a sentence that rejects a previous statement by repeating it with an added negative marker. For instance, B’s utterance in (22) is acceptable as an echo denial since it preserves the lexical choices of A’s utterance, including the use of *any time now*.

- (21) Isn’t the train going to arrive any time now?
- (22) A: Don’t worry. The train is going to arrive any time now.
B: No, it’s not going to arrive any time now. The display board says that it’s going to be late.

Thus, we predict that an immediate future PPI can co-occur with negation in “rescuing” contexts. This prediction is borne out in the case of Palestinian Arabic *da-*, as illustrated by the contrast between a declarative (23-a) and a question (23-b). This provides supporting evidence that *da-* is a PPI.

- (23) Palestinian Arabic, Abu Ghosh dialect (Afroasiatic, Semitic; Nadine Abdel-Rahman, p.c.)
- a. ?el-kahwa de-ʃ txalles^ʃ.
DEF-coffee want.PRS.3SG.M-NEG run.OUT.IPFV
(Intended: ‘The coffee is not about to run out.’)
- b. meʃ ʃallehsa:b el-kahwa da-txalles^ʃ?
NEG on.the.account DEF-coffee want.PRS.3SG.M-run.OUT.IPFV
‘Isn’t the coffee about to run out?’

Imminence PPIs are a subtype of emphatic PPIs.¹³ Emphatic PPIs conventionally require the assertion to be stronger than its alternatives (Israel 1996).¹⁴ In other words, an emphatic PPI can only occur in contexts in which it makes the assertion more informative. In negative contexts, an emphatic PPI would make the assertion weaker than its alternatives because negation reverses entailment relations. Consequently, using an emphatic PPI in a negative sentence conflicts with the conventional discourse function of the PPI, and that makes the sentence infelicitous.

For instance, (18-a), repeated here as (24-a), contains the PPI *any time now*. This sentence can be paraphrased as (25-a). Hence, the literal meaning of (24-b) should be along the lines of (25-b). Yet, (24-b) is odd because it is part of the conventional discourse function of *any time now* (but not of *in the next few minutes*) that it strengthens the assertion—a requirement that is not met in negative contexts.

- (24) a. The train is going to arrive any time now.
 b. *The train isn't going to arrive any time now.
- (25) a. The train will arrive in the next few minutes.
 b. The train won't arrive in the next few minutes.

More formally, polarity items introduce alternatives (Krifka 1995). Emphatic polarity items are conventionally associated with an *even* implication—i.e., they convey that the assertion is the least likely (i.e., strongest) alternative in the alternative set (Krifka 1995; Lahiri 1998; Crnić 2019). The *even* implication is derived by exhaustifying the statement by a phonologically null operator E, articulated in (26), whose meaning is akin to the focus particle *even* (Chierchia 2006, 2013).

- (26) *even*-exhaustification (Chierchia 2013: 148)
 $E_{ALT}(p) = p \wedge \forall q \in ALT[p <_{\mu} q]$
 Where ' $p <_{\mu} q$ ' means that p is less likely than q with respect to some contextually relevant probability measure μ .

Nicolae (2012) proposes that the polarity sensitivity of the English PPI *some* arises from *even*-exhaustification. In (27), the sentence asserts that John didn't see any-

¹³ Broadly speaking, there are two types of PPIs in terms of their discourse function: emphatic PPIs and attenuating (understating) PPIs. In contrast to emphatic PPIs, attenuating PPIs such as *sooner or later* (i) conventionally require the assertion to be weaker than its alternatives (Israel 1996). We focus on emphatic PPIs because immediate futures belong to this category.

(i) a. The train is going to arrive sooner or later.
 b. ?The train isn't going to arrive sooner or later.

¹⁴ The association between lexical meaning and polarity sensitivity is conventional rather than deterministic. For instance, *any time now* is a PPI (18), but *any time soon* is an NPI (i), even though their literal meaning is very similar. In the taxonomy of Israel (1996), *any time soon* is considered an attenuating NPI.

(i) a. *The train is going to arrive any time soon.
 b. The train isn't going to arrive any time soon.

one given a contextually-supplied domain of quantification D . By assumption, *some* activates super-domain alternatives, which are supersets of D . Crucially, these alternatives entail the assertion in a negative sentence like (27) since not seeing anyone in D' entails not seeing anyone in any subset of D' . Therefore, exhaustifying these alternatives by E leads to a contradiction: the alternatives cannot be less likely than the assertion given that they are strictly stronger than it. This inconsistency accounts for the unacceptability of (27).

- (27) *John didn't see someone.
- a. Assertion: $\neg\exists x \in D[\text{see}(\text{John}, x)]$
 - b. Alternatives: $\{\neg\exists x \in D'[\text{see}(\text{John}, x)] : D \subset D'\}$
 - c. E [John didn't see someone] =
 $\neg\exists x \in D[\text{see}(J, x)] \wedge \forall D'. D \subset D' (\neg\exists x \in D[\text{see}(J, x)]) <_{\mu}$
 $(\neg\exists x \in D'[\text{see}(J, x)])$

In what follows, we propose an exhaustification-based analysis for imminence PPIs like *in a jiffy* and immediate future markers. The phrase *a jiffy* denotes a brief time interval. Thus, *in a jiffy* is a mirror image of emphatic NPIs such as *in weeks* (29), which denote long time intervals.

- (28) a. Mary will be here in a jiffy.
b. *Mary won't be here in a jiffy.
- (29) a. Joe hasn't met Mary in weeks.
b. *Joe has met Mary in weeks.

Building on Chierchia (2013: 217-223), Iatridou & Zeijlstra (2021) suggest that *in weeks* introduces subdomain alternatives. Example (30) asserts that there is an event of Joe meeting Mary whose running time is within the time span τ . The time span τ is defined by its right boundary (RB) and left boundary (LB). In (30), the present perfect locates the RB at the utterance time (UT). The LB is modified by *in weeks*, so the LB is found weeks before the RB. Therefore, (30) asserts that Joe met Mary some time within the time span τ . The relevant alternatives are those where the meeting took place in a subinterval of τ . Importantly, all the alternatives asymmetrically entail the assertion. These alternatives need to be exhaustified by an O operator, which negates all alternatives that are not weaker than the assertion.¹⁵ However, exhaustification

¹⁵ Chierchia (2013: 143) distinguishes between emphatic NPIs, such as minimizers like *lift a finger*, and non-emphatic NPIs, such as indefinites like *any* and strong NPIs like *in weeks* (although the latter are also subject to an additional constraint, see Chierchia 2013: 213-227). According to his analysis, emphatic NPIs activate alternatives exhaustified by E (26), whereas non-emphatic NPIs activate alternatives exhaustified by O, defined in (i).

- (i) *only*-exhaustification (Chierchia 2013: 419)
 $O_{ALT}(p) = p \wedge \forall q \in ALT[p \not\subseteq q \rightarrow \neg q]$
'The assertion p is true and any alternative q not entailed by p is false.'

This approach accounts for the different discourse functions of polarity items. However, in situations where the assertion is asymmetrically entailed by its alternatives, both *even*-exhaustification and *only*-exhaustification yield a contradiction. Thus, while Nicolae (2012) proposes that the PPI *some*

leads to a contradiction: if Joe didn't meet Mary in any subinterval of τ , it cannot be the case that he met her in τ . As a result, the sentence is unacceptable.

(30) *Joe has met Mary in weeks.

- a. Assertion: $\exists e.[\text{meet}(e, \text{Joe}, \text{Mary}) \wedge \text{Run}(e) \subseteq \tau]$
- b. Alternatives: $\{\exists e.[\text{meet}(e, \text{Joe}, \text{Mary}) \wedge \text{Run}(e) \subseteq \tau'] : \tau' \subset \tau\}$

We propose a similar analysis for imminence PPIs. We define a measure function that relates time intervals to degrees representing their duration (31).

$$(31) \quad \mu_{\text{DURATION}} = \lambda t. \lambda t'. \mu_{\text{DURATION}}(t, t')$$

Example (32) asserts that there is a future time t' such that Mary is here at t' , and the duration of the interval starting from the UT (corresponding to t in (32)) and extending to t' is at most a jiffy.¹⁶ Given that *a jiffy* denotes a short time interval, the sentence evokes superdomain alternatives with intervals whose duration is more than a jiffy. For instance, one alternative can say that Mary will be here in two jiffies. In this case, the assertion asymmetrically entails all the alternatives: if Mary will be here in at most a jiffy, it follows that she will be here in at most two jiffies, and so forth. Therefore, exhaustifying the alternatives by an E operator yields a consistent meaning: the assertion is the least likely alternative in the set. Thus, the sentence is acceptable in affirmative contexts. On the other hand, negating this sentence, as in (28-b), results in a contradiction since the alternatives then asymmetrically entail the assertion.

(32) Mary will be here in a jiffy.

- a. Assertion: $\exists t'[t' > t \wedge \text{Mary is here}(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \leq a \text{ jiffy}]$
- b. Alternatives: $\{\exists t'[t' > t \wedge \text{Mary is here}(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \leq d'] : d' > a \text{ jiffy}\}$

We can now turn to immediate future markers. Like lexical imminence PPIs, immediate futures are typically vague about the degree of temporal proximity. An immediate future like *be about to* indicates that the event in question is predicted to occur in the near future, but it does not pose any strict limitations on what is considered to be the near future. Moreover, temporal proximity is context-dependent. For instance, (33) would probably be interpreted as predicting that it will start raining in the next few hours. In contrast, (34) refers to an event that is predicted to happen 100 years from now but is perceived as proximate in the context of the future of the planet or humanity.

is exhaustified by E, Spector (2014); Nicolae (2017) derive the PPI-hood of disjunctions like French *ou* 'or' and *soit... soit* 'either... or' by means of *only*-exhaustification, instead of *even*-exhaustification. Importantly, both analyses make the same predictions. Nevertheless, we prefer to use *even*-exhaustification to capture the emphatic value of imminence PPIs as end-of-scale elements, although our analysis does not depend on this choice.

¹⁶ English *in*-adverbials usually have an exhaustive ('exactly') interpretation, but they can also have just an upper-bounded interpretation (Mittwoch 2009: 260-261), e.g., *You'll be back on your feet in two weeks' time*.

- (33) It's about to rain.
- (34) Climate change is about to transform Earth into an unrecognizable, alien landscape.¹⁷

Following Bochnak & Klecha (2018), we analyze vague tenses in a similar fashion to the positive form of relative adjectives like *tall*.¹⁸ Specifically, an immediate future implies that the reference time (RT) is close to the evaluation time, which is typically the UT in main clauses. The notion of closeness is based on the duration measure function μ_{DURATION} . The degree scale S associated with the duration measure function is divided into three parts, whose boundaries are contextually determined: a range of degrees L_{DURATION}^- consisting of degrees that would be considered short given a context c , a neutral range N_{DURATION} of degrees that are neither short nor long, and a range of degrees L_{DURATION}^+ that would be considered long.¹⁹ We propose the denotation in (36) for the immediate future, where t' corresponds to the RT, and t corresponds to the evaluation time. For reference, we assume the denotation in (35) for the simple future.

$$(35) \quad \llbracket \text{FUT} \rrbracket = \lambda P_{\langle i, st \rangle} \lambda t \lambda w. \exists t' [t' > t \wedge P(t')(w)]$$

$$(36) \quad \llbracket \text{IMM.FUT} \rrbracket = \lambda P_{\langle i, st \rangle} \lambda t \lambda w. \exists t' [t' > t \wedge P(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \in L_{\text{DURATION}}^-]$$

A statement containing a PPI immediate future obligatorily activates alternatives and is exhaustified by an E operator. For instance, (37) asserts that it will rain in the near future. The relevant alternatives are those where the duration of the interval between the evaluation time and the reference time is within a range that properly contains L_{DURATION}^- . In other words, we consider ranges that contain L_{DURATION}^- but also overlap with N_{DURATION} , and in some cases even with L_{DURATION}^+ . In affirmative (upward-entailing) contexts, the assertion entails all the alternatives (similarly to example (32) with *in a jiffy*), and exhaustification by an E operator yields consistent and meaningful results. Thus, (37) means that it will rain in the near future and that it is less likely that it will rain in the near future than that it will rain some time within the near or not-so-near future.

¹⁷ Osborne, Hannah. *Newsweek*, 3 September 2018, <https://www.newsweek.com/earth-major-transformation-climate-change-global-warming-1101857>

¹⁸ Bochnak & Klecha (2018) focus on vagueness in past time reference. A point of difference between their analysis and ours is that they assume a pronominal analysis of tense, while our denotations in (35) and (36) assume a quantificational analysis. However, the debate between these two views of tense is tangential to our discussion. Furthermore, there is a common view that future is not a proper tense but a modal or aspectual category (see e.g., Copley 2009). Since this issue is not relevant to our analysis either, we will treat future as a regular tense for the sake of simplicity and brevity.

¹⁹ von Stechow (2009) proposes a denotation of the positive operator that directly refers to the neutral range. Since we are not suggesting that an immediate future is gradable, we do not decompose it into a gradable tense and a positive operator, in contrast to the way gradable adjectives are often analyzed. Instead, we build the contribution of the positive operator into the denotation of the immediate future marker.

(37) It IMM.FUT to rain.

- a. Assertion: $\exists t'[t' > t \wedge \text{rain}(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \in L_{\text{DURATION}}^-]$
- b. Alternatives: $\{\exists t'[t' > t \wedge \text{rain}(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \in L'_{\text{DURATION}}] : L'_{\text{DURATION}} \supset L_{\text{DURATION}}^- \}$

By contrast, in negative contexts, exhaustification by E yields a contradiction. Example (38) means that it will not rain in the near future and that it is less likely that it will not rain in the near future than that it will not rain in the near or not-so-near future. Given that the assertion is entailed by the alternatives, it cannot be less likely. This results in unacceptability.

(38) It NEG IMM.FUT to rain.

- a. Assertion: $\neg \exists t'[t' > t \wedge \text{rain}(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \in L_{\text{DURATION}}^-]$
- b. Alternatives: $\{\neg \exists t'[t' > t \wedge \text{rain}(t')(w) \wedge \mu_{\text{DURATION}}(t, t') \in L'_{\text{DURATION}}] : L'_{\text{DURATION}} \supset L_{\text{DURATION}}^- \}$

In summary, we propose that a immediate future asserts that the duration of the interval between the evaluation time (typically the UT) and the RT is short with respect to the context. Regarding the observation that immediate futures are often PPIs, we argue that these markers activate superdomain alternatives where the duration of the interval between the evaluation time and the RT is not necessarily short. These alternatives are exhausted by an E operator, producing a consistent meaning in affirmative contexts but a contradiction in negative contexts, where all the alternatives entail the assertion. This accounts for the PPI-hood often exhibited by immediate futures.

4 Back to symmetry: Loss of polarity sensitivity

In Section 2, we proposed a diachronic pathway of development of paradigmatic symmetry in the context of future markers. A future marker that goes through the three stages starts out as a PPI and eventually loses its polarity sensitivity. In this section, we speculate on the potential causes and mechanisms behind this diachronic change. By doing so, we aim to illustrate that our analysis of immediate futures as PPIs, presentend in Section 3, aligns with the trajectory in (6). According to our analysis, a novel future marker is a PPI if (i) it is an immediate future; and (ii) it is conventionally associated with an *even* implication. Hence, we predict that a future marker will lose its polarity sensitivity if it no longer meets these two conditions.

One scenario is that the future marker undergoes semantic bleaching and develops into a simple future. Thus, the future marker is no longer stronger than its alternatives. According to Bybee et al. (1994: 271), there is no sufficient historical evidence that immediate futures tend to develop into simple futures.²⁰ However, they argue that

²⁰ Bybee et al. (1994: 271) cite Fleischman (1983: 193) as providing supporting evidence for this diachronic pathway. However, to be precise, Fleischman argues that *go*-constructions in Romance

immediate futures are typically younger than simple futures, which suggests that this pathway is plausible. Moreover, Grossman et al. (2014: 92–93) show that the Coptic future tense, which does not show remoteness distinctions, originates in a construction whose earliest attestations are associated with immediate futurity. We speculate that such a diachronic change might have happened in certain dialects of Palestinian Arabic. As described in Subsection 2.1, the immediate future *da-* is a PPI in the Abu Ghosh dialect (9). However, in the neighboring Jerusalem dialect, *da-* is a simple future, and it can be negated (39). It is possible that the Jerusalem dialect exhibits a later stage in the development of this future marker. In that case, this marker might have been a PPI in the Jerusalem dialect as well, but it later lost its polarity sensitivity when it developed from an immediate future into a simple future.

- (39) Palestinian Arabic, Jerusalem dialect (Afroasiatic, Semitic; Rasha Abdeen, p.c.)
- a. *da-tmat^fr.*
want.PRS.3SG.M-rain.IPFV
'It's going to rain.'
 - b. *de-ʃ tmat^fr.*
want.PRS.3SG.M-NEG rain.IPFV
'It's not going to rain.'

Another scenario is that a future marker loses its polarity sensitivity while maintaining its semantics as an immediate future. Hypothetically, a future marker can lose its conventionalized *even* implication without any further semantic change. This kind of process is commonly observed in grammaticalization: emphatic expressions tend to lose their emphatic value over time (Dahl 2001; Haspelmath 1999). For instance, emphatic negation markers often become ordinary negation markers through a Jespersen's cycle. According to Dahl (2001) and Haspelmath (1999), speakers introduce innovations such as novel emphatic expressions to attract the attention of their listeners. If an innovation is picked up by other speakers and becomes widespread within the speech community, it turns more frequent. Overused expressions tend to lose their emphatic value due to habituation.²¹

We are not aware of a clear case where an immediate future construction lost its polarity sensitivity without becoming a simple future. It is worth noting, however, that immediate futures are not all PPIs. In present-day English, *be about to* can be negated in its immediate future use. For example, (40) means that Bhutan's policy of admitting few tourists will not change in the near future, but it does not say that this policy will remain in place forever. Being an immediate future, *be about to* is incompatible with *never* (41), which may be considered a diagnostic for an immediate future since the meaning of *never* conflicts with the temporal restriction to the near

languages and English had a "present relevance" requirement in the early stages of their grammaticalization, and this requirement was relaxed as these constructions developed into simple futures.

²¹ Danckaert & Gianollo (2024) [this volume] examine a similar case of rhetorical overuse, arguing that Latin *quidem* 'indeed' not only lost its original discourse requirement of shared knowledge due to overuse but also acquired new uses in negative and adversative contexts.

future. It remains to be seen whether *be about to* and other immediate futures that may co-occur with negation were PPIs in earlier stages of their grammaticalization.

- (40) Bhutan has comfortably courted a niche tourism market since 1974. In 2011 it had 37,481 tourists, 2,795 of them from Britain. “High-value, low-volume” is the official mantra. That is not about to change, though more tourists and their income would be welcome. [iWeb]
- (41) #That is never about to change.

Another potential contributing factor to the development of paradigmatic symmetry is that some PPI future markers are compatible with negation when they are used in their original lexical sense, as in the case of *laik* ‘want / be about to’ in Tok Pisin (8). The existence of such instances may allow the extension (in the sense of Harris & Campbell 1995) of negation to the future use of these expressions.

5 Result-oriented approaches to paradigmatic asymmetry

Several explanations have been proposed in the literature for paradigmatic asymmetry, which can be characterized as result-oriented or source-oriented (Cristofaro 2017, 2019; Schmidtke-Bode 2019). Result-oriented explanations in typology attribute language universals to attractor states that languages are drawn into as they develop. By contrast, a source-oriented explanation attributes a recurrent cross-linguistic pattern to the actual diachronic processes that produced this pattern rather than to principles of grammar or usage that favor one synchronic state over another. However, as Schmidtke-Bode & Grossman (2019: 234) point out, these two modes of explanation are not mutually exclusive.

One result-oriented explanation for paradigmatic asymmetry that has been proposed is that certain tense-aspect markers are absent in the negative because of the stativizing effect of negation (Schmid 1980; Matthews 1990). According to this view, negated predicates denote unrealized, and hence uncompleted, events. Therefore, negation tends to be incompatible with aspectual markers that refer to completed events such as perfectives, while imperfectives are predicted to be compatible with negation. However, Miestamo & van der Auwera (2011: 68-70) observe that perfectives and imperfectives (broadly defined) are equally often blocked from negative contexts. In other words, it is just as common for a language to have an imperfective that is incompatible with negation, even though such markers are used to refer to uncompleted events. Furthermore, in most languages, perfective and imperfective markers are fully compatible with negation, so paradigmatic asymmetry is in fact the exception rather than the rule.

Another result-oriented explanation posits that the development of paradigmatic asymmetry is motivated by a pressure for economical coding of grammatical categories (Miestamo 2005: 200-202). According to (Givón 1978: 80), negative clauses are typically used to reject a proposition that is salient in the discourse context,

whether because it has been previously discussed or because the speaker assumes that the addressee believes it to be true. For instance, a sentence like *My name isn't Bob* is odd if the addressee has no particular reason to assume that the speaker's name is Bob. By contrast, affirmative clauses typically do not require the corresponding negative to be discourse salient. Consequently, when asserting a negative sentence, the communicative benefit of providing explicit information about the event under discussion (i.e., who, when, and where?) is relatively small since this information is already contextually given due to the salience of the corresponding affirmative. Generally, speakers do not provide more information than is required by current conversational purposes (the second part of Grice's Maxim of Quantity (Grice 1989)). Therefore, a cooperative speaker is more likely to provide information about the time, place, and participants of an event in the affirmative than in the negative. According to Miestamo (2005: 211), this usage preference is conventionalized in some languages as a restriction of certain grammatical markers—e.g., tense, aspect, or person—to the affirmative. This leads to paradigmatic asymmetry, where certain grammatical distinctions are neutralized in the negative. While this approach suggests a functional rationale for paradigmatic asymmetry, it still remains to be explained why in a given language, negation is incompatible with a specific TAM marker, yet compatible with another marker that stands in paradigmatic opposition to it. To this end, Miestamo (2005: 214) and Miestamo & van der Auwera (2011: 77-82) propose that the form that survives under negation is the one that is formally and semantically unmarked.

In contrast to these result-oriented approaches, we follow the source-oriented approach advocated by Givón (1978: 97-99). Notice that the diachronic pathway proposed by Givón (See Section 1) may also explain the cross-linguistic tendency that when a language exhibits paradigmatic asymmetry, the form that can be negated is usually the shorter one (Miestamo & van der Auwera 2011: 77-82). Older grammatical markers tend to be shorter than novel markers since grammaticalization typically involves phonological reduction (Bybee et al. 1994: 6). Hence, the shorter form is compatible with negation not because it is the unmarked form (as argued by Miestamo (2005: 214) and Miestamo & van der Auwera (2011: 77-82)) but because it is older and thus had more time to spread to negative contexts. In conclusion, while we acknowledge the insights of result-oriented approaches to understanding paradigmatic asymmetry, our analysis itself does not attribute a functional purpose to these asymmetries.

6 Conclusion

In this paper, we have proposed a diachronic explanation for paradigmatic asymmetry in future marking. It is plausible that other cases of TAM markers occurring only in affirmative contexts can be accounted for in a similar way. For instance, the Spanish recent past construction *acabar de* V.INF is banned from negative contexts (42).²² A

²² Note that Spanish *acabar de* V.INF is compatible with negation when used in a modal sense, e.g., (i), or as a recent past in echo denial contexts (Martínez-Atienza 2017). Interestingly, the French

recent past produces a stronger statement than a general past in affirmative contexts. Hence, if a recent past is conventionally associated with an *even* implication, it will be a PPI.²³

(42) Spanish

- a. Juan acaba de comprar un coche.
 PN finish.PRS.3SG of buy.INF a car
 ‘Juan just bought a car.’ (Rosemeyer & Grossman 2017: 517)
- b. ?Juan no acaba de comprar un coche.
 PN NEG finish.PRS.3SG of buy.INF a car
 (Intended: ‘Juan didn’t just buy a car.’) (Malte Rosemeyer, p.c.)

To give another example, the Persian progressive construction *dār-ad* V.IPFV cannot be negated (43). In this case, too, an affirmative statement with the progressive asymmetrically entails the corresponding statement with the imperfective (Deo 2009, 2015). Therefore, progressive constructions—especially novel ones—are potential PPIs too. We leave it for future research to further develop this analysis.

(43) Persian (Nematollahi 2018: 144-145)

- a. sām dār-ad mi-āy-ad.
 PN have.PRS-3SG IPFV-COME.PRS-3SG
 ‘Sām is coming.’
- b. *sām dār-ad ne-mi-āy-ad.
 PN have.PRS-3SG NEG-IPFV-COME.PRS-3SG
 (Intended: ‘Sām isn’t coming.’)

In proposing a diachronic, source-oriented explanation for synchronic distributions, we follow a line of thought that goes back at least to Greenberg (1969, 1978) and has been argued for explicitly by Givón (1979), Bybee (1988, 2006), Blevins (2004), Cristofaro (2012), and others. In this view, synchronic distributions are the product of processes of change. At the very least, this means that “any synchronic pattern must have a diachronic dimension, since that pattern had to come into being in some way” (Bybee 2006: 179).

recent past *venir de* V.INF can be negated, e.g., (ii). It is possible that the compatibility with negation is a late development in the grammaticalization of *venir de* V.INF.

- (i) No acabo de entender el motivo de tu enfado.
 NEG finish.PRS.1SG of understand.INF the reason of your anger
 ‘I don’t quite understand the reason for your anger.’ (Martínez-Atienza 2017)
- (ii) Je (ne) viens pas de voir Charles.
 I NEG come.PRS.1SG NEG of see.INF PN
 ‘I haven’t just seen Charles.’ (Stéphane Polis, p.c.)

²³ It remains to be seen how this proposal fits with that made in (Rosemeyer & Grossman 2017), according to which early Spanish *acabar de* V.INF had a very particular discourse function associated with informativity.

Of course, this pushes the ‘black box’ of explanation around a bit, but does not eliminate it, since the question remains: what are the motivations for language change? Answers to this question are diverse, but tend to gravitate to explanations based on recurrent patterns of usage, such as frequency or discourse profile, on constraints on linguistic representations, or on constraints on acquisition or learnability (Haspelmath 2019; Schmidtke-Bode & Grossman 2019). The explanation offered in this paper makes reference to both usage-based and representational factors.

In this paper, we have made several proposals about the diachronic pathways that lead to paradigmatic asymmetry and symmetry in future marking. One major claim is that tense markers characterized by remoteness distinctions tend to be positive polarity items. Another set of proposals deals with the processes through which future markers might lose their polarity sensitivity, thereby paving the way for paradigmatic symmetry to develop. Finally, we have suggested that the analysis offered here might be relevant for other grammaticalization pathways, such as recent pasts or progressives. However, it is clear that much empirical and theoretical work remains to clarify the robustness of these proposals.

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Abbreviations

AGT agentive case; ACT actual; ADEL adelative; ANOM agentive nominalizer; AUX auxiliary; DEF definite; ERG ergative; F feminine; FUT future; IMPERS impersonal; INCH inchoative; INF infinitive; INDEF indefinite; IPFV imperfective; IRR irrealis; M masculine; NEG negation; NF near future; PERS personal; PL plural; PM predicate marker; PN proper name; POT potential; PRF perfect; PROB probable; PROSP prospective; PRS present; PST past; RF remote future; SG singular.

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